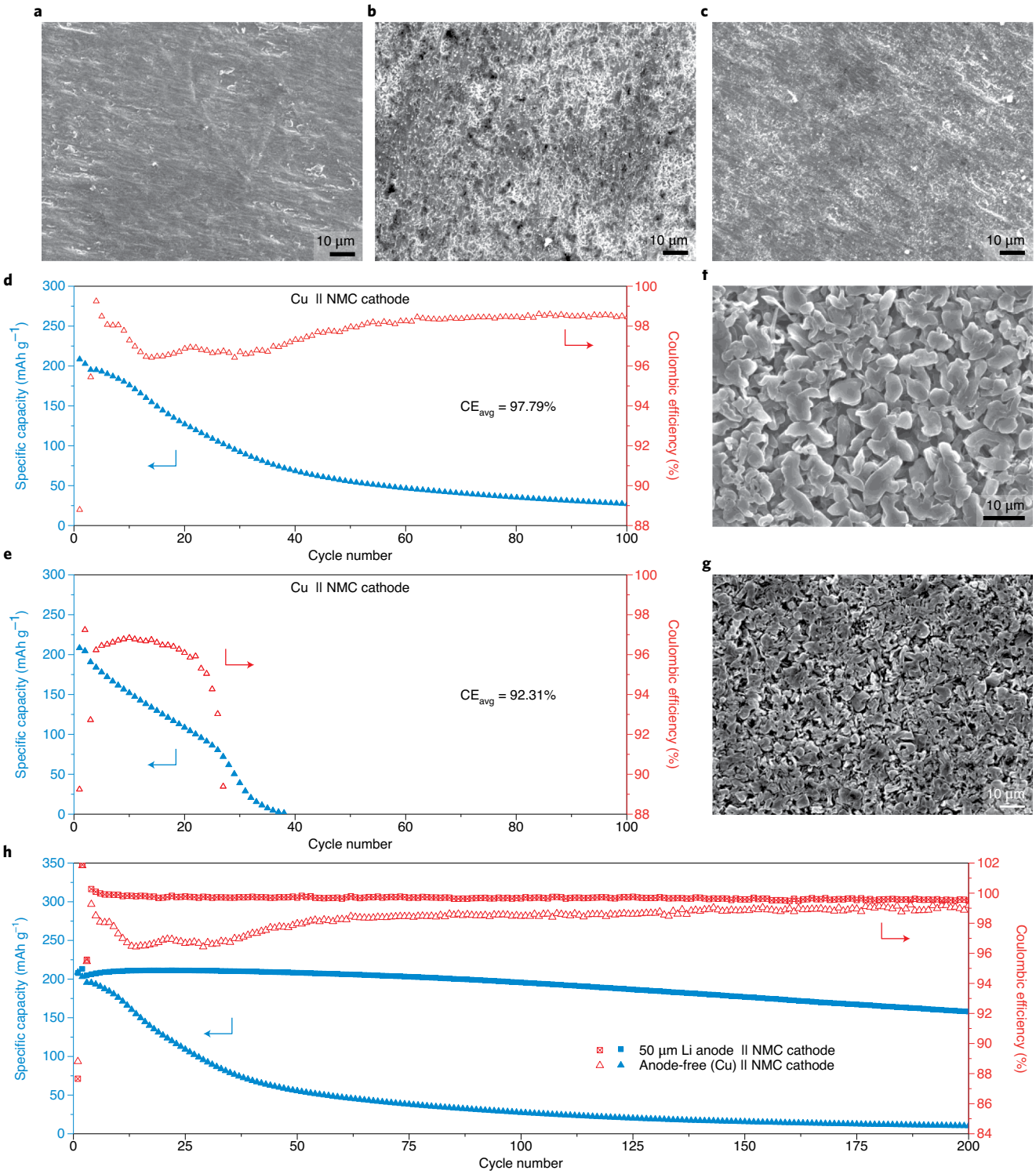

Fig. 2 | Impacts from Cu substrates, electrolytes and Li on measured CE and observed cycling stability of anode-free NMC batteries. a, Morphology of bare Cu. **b**, Li deposited (0.1 mAh cm^{-2}) on Cu in compatible electrolyte comprised of 1.2 M LiFSI in TEP/BTFE (1:3 by molar ratio), which indicates that Li has preferred nucleation sites on Cu. **c**, Remaining Li of **b** after Li being stripped suggesting that 'dead' Li readily formed even after the first cycle. **d**,**e**, Cycling performance and CE of anode-free Cu/NMC811 coin cells in compatible electrolyte: 1.2 M LiFSI in TEP/BTFE (**d**) and conventional carbonate electrolyte: 1.0 M $LiPF_6$ in EC/EMC-2 wt% VC (**e**). **f**,**g**, Morphologies of Li deposited on Cu at depositing capacity of 4.0 mAh cm^{-2} in different electrolytes. Although anode-free cells are used, Li particles formed in compatible electrolyte are still larger than those in carbonate electrolytes thus less amount of side reactions are happening on newly exposed Li surfaces decelerating side reactions during each cycle. **h**, Comparison of cycling performance and CE for 50-μm-Li anode/NMC811 and anode-free Cu/NMC811 coin cells, where the cells after two formation cycles were charged at C/10 to 4.4 V, and then discharged at C/3 to 2.7 V. The addition of 50 μm Li in the anode-free cells significantly boosts the stable cycling and CE.

Another observation in Fig. 2d,e is that CE has no direct relationship with usable capacity. CE can be very high regardless the deliverable capacity is 100% or 50% of its original capacities. The reason is that both the numerator and denominator of equation (1) are changing in each cycle.

Impacts of original Li content in the cell. In 'thin film' cells with 'flooded' electrolytes and 'unlimited' Li metal anode, an artificially long cycling can still be demonstrated even if a 'bad' electrolyte is chosen[3]. When the electrolyte is more than enough, a fast degradation of the anode-free cell is usually caused by fast SEI build-up (cell impedance increase) and/or the quick consumption of Li[16] that converts into 'dead' Li wrapped by thickened SEI layers. If additional Li is provided in the beginning on the anode, CE and cycle life will be readily improved as long as the cell impedance is still not high enough to terminate the cell. For example, the averaged CE in Cu/NMC cell is only 97.79% (Fig. 2h), while that averaged CE of Li/NMC is 99.69% in the same electrolyte. It confirms that an excessive amount of Li in the cell plays a large role in determining the CE and thus cycling stability of a Li metal cell, provided that electrolyte is always sufficient. A recent publication on the quantification of 'dead' Li after cycling[17] provides interesting insights. However, a non-destructive method to quantify residual Li and electrolyte during each cycle without disassembling the cell is still urgently called for.

It is noted that for Li/NMC cell in Fig. 2h, a small capacity increase is displayed for the first ten cycles, while anode-free NMC cell displays continuous capacity fading from the very beginning. The reason is that NMC cathode used here is not only high mass loading (23 mg cm^{-2}) but also dense (porosity: 30%). It is not uncommon to see slight capacity increase in the initial cycles due to the improved electrolyte wetting within the dense cathode. For anode-free NMC cells, however, the discharge capacity is directly related to the amount of Li^+ ions available in the cell which is unfortunately being lost from the very beginning[16,18].

Cathode/electrolyte interface. The choice of cathode also impacts CE of the anode-free cells, although CEI impedance usually does not dominate cell failure as long as the cut-off voltage is lower than 4.3 V[19,20]. Sometimes, a higher cut-off voltage at 4.4 V (versus Li^+/Li) is necessary to extract more energy. Electrolyte decomposition and CEI thickening are therefore unavoidable at high voltages, 'trapping' Li^+ ions on the cathode. After extensive cycling, the impacts from CEI impedance will be displayed by accelerating cell failure. If a low cut-off voltage cathode is used at the very beginning for example, by using $LiFePO_4$ (LFP) cathode, the narrowed electrochemical window (up to 3.8 V) will lead to a higher CE and a longer stable cycling as compared to anode-free NMC cells[21–23].

Li/Cu cells

Unlike in a Li/NMC or anode free cells where CE is calculated based on cathode capacity, CE in a Li/Cu cell (Fig. 1g) is calculated based on the ratio between the amount of stripped Li from the Cu substrate and deposited Li onto the Cu substrate, as described in equation (3). We can also rewrite the equation based on the electrons flowing into or out of the Cu current collector. When plating/stripping Li at low potentials, irreversible electrochemical reactions very often count for an enlarged portion in the denominator and/or the numerator in equation (3). If the amounts of electrons from irreversible electrochemical reactions become substantial, CE measured will not be able to provide quantifiable information of Li^+ ions left in the cell.

$$\mathrm{CE} = \frac{\text{Stripping capacity}}{\text{Depositing capacity}} = \frac{\text{Number of e}^- \text{ from stripping Li}^+ + \text{Number of e}^- \text{ from irreversible electochemical reactions}}{\text{Number of e}^- \text{ from depositing Li}^+ + \text{Number of e}^- \text{ from irreversible electochemical reactions}} \quad (3)$$

A high CE measured in a Li/Cu cell, even with an appropriate electrolyte, cannot guarantee the successful application of the electrolyte in a Li/NMC or anode-free cell. One of the main reasons is that in Li/Cu cells the voltage used for stripping Li is usually less than 1.5 V, avoiding the issues associated with high voltages. However, the CE measured in Li/Cu cells may be very close to those measured in anode-free cells with a LFP cathode given that most electrolytes are stable below 3.8 V[21].

Different protocols and methods have been suggested for Li/Cu cells[24–26], but CE measured, no matter in which way, is not directly relevant to the full cells. It can be used to compare the properties of different electrolytes at low potentials, provided that the testing conditions for example, capacity, rate, electrolyte amount and Li thickness are the same in all tests. For example, the electrolyte amount also significantly impacts the electrochemical properties of a Li/Cu cell. In Fig. 3a, the averaged CE in Li/Cu cell with 75 μL electrolyte is 99.54%, higher than that of the same cell tested with 15 μL of the same electrolyte (CE: 98.60%). The polarization of Li/Cu cells with lean electrolyte also increases more rapidly than those with flooded electrolyte during cycling. The fast depletion of lean electrolyte leads to high polarization, high cell impedance and low CE. Accordingly, cell capacity degrades fast in lean electrolyte, although dense Li is still available. The similar impact of electrolyte content can also be found in Li/NMC cells (Fig. 3b).

It is also important to control the amount of Li deposition in Li/Cu cell testing to obtain meaningful results of cycling. In Li/Cu cell, a fixed amount of Li is deposited on Cu substrate through capacity/time control[24,26]. If the deposited Li on Cu is controlled at 1 mAh cm^{-2} during each cycle while Li source on the Li metal side is as thick as 460 μm (= 92 mAh cm^{-2}), then only 1/92 ≈ 1% of Li anode is being utilized each cycle. As long as there is sufficient Li and electrolyte in the cell, an artificially ultra-long cycling can be achieved. If the goal is to build a high-energy Li metal cell, we recommend that the deposited Li on Cu needs to be comparable with the loading of cathode to be coupled in a full cell, that is, 4 mAh cm^{-2} (ref. [2,3]). The similar criteria need to be considered when evaluating Mg, Zn or Na batteries.

Li/Li cells

Symmetric Li/Li cells (Fig. 1h) are sometimes used to evaluate electrolyte or Li metal properties but it is not quite meaningful. This is because Li/Li cells use capacity/time control for both Li deposition and stripping, so CE is always 100%. However, Li/Li symmetrical cells may be used to assess separator membrane[27] or soild state electrolyte membranes[15] in terms of its capability to induce or mitigate rapid dendrite growth and penetration. But the amount of deposited/stripped Li of each cycle needs to be relevant to the practical applications. Similarly, symmetrical Mg/Mg or Zn/Zn cell does not provide sufficient information to fully evaluate new concepts neither.

Protocol for measuring and applying CE in LMBs

We now propose a general approach of measuring CE in coin cells to assess new concepts of electrodes and electrolytes for high-energy LMBs, as illustrated in Fig. 4. While Li/Li symmetric cells provide limited information, Li/Cu cells are able to obtain some useful information to evaluate the compatibility between the lithium anode and the electrolyte. Note that Li deposition is also happening on the Li side in Li/Cu cells. This means SEI and cell impedance are also being built on Li side upon cycling. A very thick Li foil, coupled with an 'unlimited' amount of electrolyte easily demonstrates a few thousands of artificially stable cycling, which is not recommended. It is suggested that 50-μm-thick Li could be a good start in Li/Cu cells. During each cycle, plating 4 mAh cm^{-2} (≈ 20 μm) of Li on the copper side is a reasonable areal capacity to be coupled with thick cathode for high-energy LMBs[2,3]. The electrolyte does not need

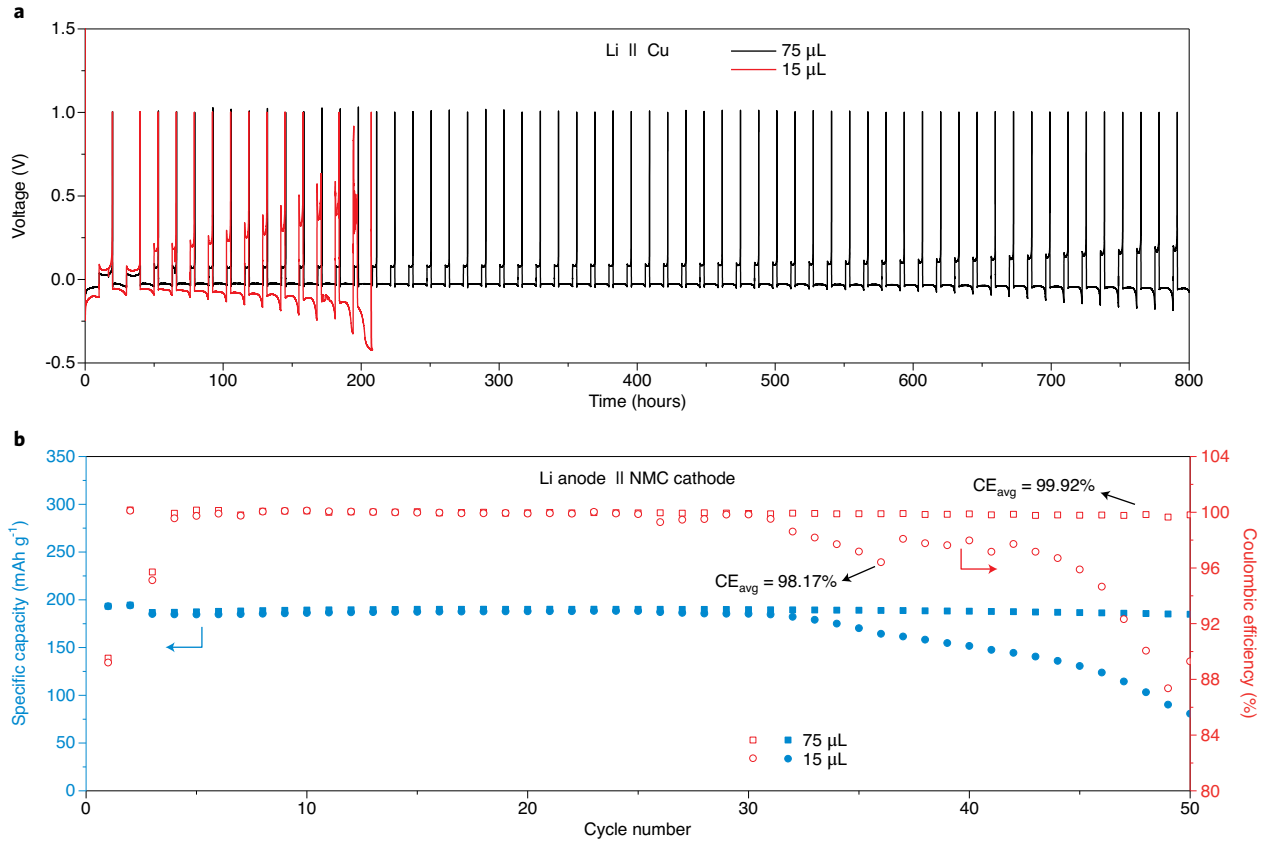

Fig. 3 | Impacts of electrolyte amounts on CE and cycling of Li||Cu and Li||NMC cells. a, Voltage profile of 50 μm Li/Cu coin cells with different amounts of compatible electrolyte comprised of 1.2 M LiFSI in TEP/BTFE (1:3 by molar ratio). Here the Li depositing amount is 4 mAh cm^{-2} and the cut-off voltage of stripping is 1.0 V. After two formation cycles at current density of 0.4 mA cm^{-2}, coin cells were cycled by depositing at current density of 0.4 mA cm^{-2} and stripping at current density of 1.32 mA cm^{-2} up to 1.0 V. The more electrolyte is added in the cell, the lower polarization of each deposition/stripping process and the longer stable cycling can be achieved. **b**, Comparison of cycling performance and CE of 50 μm Li/NMC622 coin cells with different amount of electrolyte. The coin cell with more electrolyte displays higher CE and longer cycling although all the other testing conditions are kept the same. The results indicate that electrolyte content has a considerable impact on CE measurements and cell cycling and thus needs to be consistently controlled in all tests of Li metal batteries.

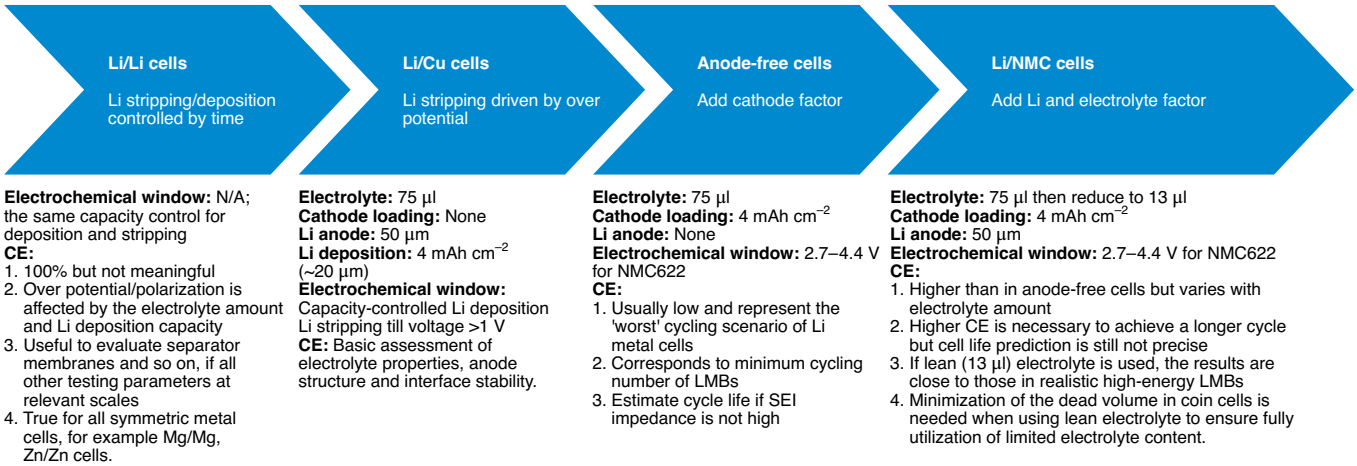

Fig. 4 | A coin cell protocol to measure and interpret CE in LMBs. The suggested coin cell parameters, that is, Li deposition capacity, Li thickness, cathode loading and lean electrolyte amount (13 μl for an electrode area of 1.27 cm^2) are based on the cell design of a realistic high energy LMB of 300 Wh kg^{-1} pouch cell[9,10]. The cathode porosity is controlled at ~35% and separator is 20 μm thick with a porosity of 45%–50%. The electrode area in the coin cell protocol can change but the ratio of lean electrolyte to capacity is ~3 g (Ah)$^{-1}$ or lower. The thickness of Li metal can be thinner than 50 μm but the actual lithium thickness or areal mass should be documented. These suggested parameters may change depending on the cell design, but the cathode loading and lean electrolyte amount will be around the numbers suggested here for designing a high energy LMB.